

2.1

Sea A una matriz columna de orden $n \times 1$ tal que $A^t A = 1$ y $B = Id_n - 2AA^t$. Demostrar que:

a) B es simétrica

b) $B^{-1} = B^t$

2.2

Sea X una matriz cuadrada de tamaño $n \times n$ y elementos reales. Sea k un número par. Probar que si $X^k = -Id$, entonces n es también un número par.

2.3

Hallar el siguiente determinante para $n \geq 2$

$$A_n = \begin{vmatrix} x_1 + y_1 & x_1 + y_2 & x_1 + y_3 & \cdots & x_1 + y_n \\ x_2 + y_1 & x_2 + y_2 & x_2 + y_3 & \cdots & x_2 + y_n \\ x_3 + y_1 & x_3 + y_2 & x_3 + y_3 & \cdots & x_3 + y_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n + y_1 & x_n + y_2 & x_n + y_3 & \cdots & x_n + y_n \end{vmatrix}$$

2.4

Calcular el siguiente determinante:

$$\begin{vmatrix} 0 & x_1 & x_2 & \cdots & x_n \\ x_1 & 1 & 0 & \cdots & 0 \\ x_2 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n & 0 & 0 & \cdots & 1 \end{vmatrix}$$

¿ Para qué valores reales de $x_1, x_2, \dots, x_n$ se anula?.

2.5

Encontrar la (única) respuesta correcta, de entre las indicadas, a las siguientes cuestiones:

Si $A \in \mathcal{M}_{n \times n}(\mathbb{R})$ entonces.

☐ A simétrica $\Rightarrow -A$ antisimétrica.

☐ A simétrica $\Rightarrow A^k$ simétrica, para cualquier $k \in \mathbb{N}$.

☐ A simétrica $\Rightarrow A$ no es antisimétrica.

☐ Ninguna de las anteriores respuestas es correcta.

2.6

Encontrar la (única) respuesta correcta, de entre las indicadas, a las siguientes cuestiones:

En una matriz cuadrada de orden n escribimos las filas en orden inverso.

- ☐ Su determinante queda multiplicado por -1 .
- ☐ Su determinante queda invariante.
- ☐ Su determinante queda multiplicado por $(-1)^n$.
- ☐ Ninguna de las restantes respuestas es correcta.

2.7

Dados $x \in \mathbb{R}$, $x \neq 0$ y la matriz

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & x & x \\ 1 & x & 0 & x \\ 1 & x & x & 0 \end{pmatrix},$$

calcular $\det(A^3)$ y $\det(A^{-1})$.

2.8

Dado $a \in \mathbb{R}$ y para cada n entero positivo definimos las matrices $A_n \in M_{n \times n}(\mathbb{R})$:

$$A_n = \begin{pmatrix} 3 & a & a & \dots & a & a \\ a & 3 & a & \dots & a & a \\ a & a & 3 & \dots & a & a \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a & a & a & \dots & 3 & a \\ a & a & a & \dots & a & 3 \end{pmatrix}$$

- (a) Calcular en función de a , $\det(A_4)$.
- (b) Calcular en función de a y n , $\det(A_n)$.

2.9

Choose the numbers a, b, c, d in this augmented matrix so that there is (a) no solution
(b) infinitely many solutions.

$$[A \quad \mathbf{b}] = \begin{bmatrix} 1 & 2 & 3 & a \\ 0 & 4 & 5 & b \\ 0 & 0 & d & c \end{bmatrix}$$

Which of the numbers a, b, c , or d have no effect on the solvability?

2.10

Suppose you solve $Ax = b$ for three special right sides b :

$$Ax_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad Ax_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad Ax_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

If the three solutions x_1, x_2, x_3 are the columns of a matrix X , what is A times X ?

If the three solutions are $x_1 = (1, 1, 1)$ and $x_2 = (0, 1, 1)$ and $x_3 = (0, 0, 1)$, solve $Ax = b$ when $b = (3, 5, 8)$. Challenge problem: What is A ?

2.11

A és una matriu 4×4 amb 1's a la diagonal principal, $-a, -b, -c$ formant la diagonal menor immediatament per sobre de la diagonal principal i amb els demés elements 0. Calculeu la inversa de A , tant pel mètode de Gauss com aplicant la regla de Cramer i comproveu que obteniu el mateix resultat.

2.12

$$\text{Let } A = \begin{bmatrix} 0 & 0 \\ 6 & 9 \\ 2 & 3 \end{bmatrix}. \quad b = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}. \quad \text{Find the complete solution to } Ax = b.$$

(Remember that the general solution is a sum of the general solution of the homogeneous system plus a particular solution)

2.13

Suppose A is a 3 by 5 matrix and the equation $Ax = b$ has a solution for every b . What are (a)(b)(c)(d)? (If you don't have enough information to answer, tell as much about the answer as you can.)

(a) Column space of A

(b) Nullspace of A

(c) Rank of A

(d) Rank of the 6 by 5 matrix $B = \begin{bmatrix} A \\ A \end{bmatrix}$

2.14

- (a) Suppose three matrices satisfy $AB = C$. If the columns of B are dependent, show that the columns of C are dependent.

- (b) If A is 5 by 3 and B is 3 by 5, show using part (a) or otherwise that $AB = I$ is impossible.

2.15

Apply row elimination to reduce this invertible matrix from A to I . Then write A^{-1} as a product of three (or more) simple matrices coming from that elimination. Multiply these matrices to find A^{-1} .

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 4 & 0 & 1 \end{bmatrix}.$$

2.16

Let $B = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 4 \\ 3 & 7 & k \end{pmatrix}$

- A) For which values of k is the matrix B singular?
 B) Find all values of k for which the system $Bx = (1, 2, 3)^T$ has infinitely many solutions.
 C) Find all values of k for which the system $Bx = (10, 1, 2014)^T$ has exactly one solution.

2.17

- (a) Suppose the matrices A and B have the same column space. Give an example where A and B have different nullspaces — or say why this is impossible.
- (b) Again A and B have the same column space. Give an example where A and B have different ranks r — or say why this is impossible.
- (c) CIRCLE TRUE or FALSE:
 If B is a square matrix then $\mathbf{C}(B) = \mathbf{C}(B^T)$.
- (d) If the columns of a 5 by 3 matrix M are linearly independent and x in $\mathbf{R}^3$ is not the zero vector, then you know that Mx is _____. I am looking for an answer that uses independence of columns and $x \neq 0$.

2.18

Sea $A \in M_{n \times n}(\mathbb{R})$ una matriz cuadrada cumpliendo $A^2 + A + Id = 0$.

- (i) Demostar que A es inversible.
 - (ii) Probar que $A^{-1} = -(A + Id) = A^2$.
 - (iii) ¿Cuánto vale A^3 ? ¿Y A^{2013} ?
-

2.19

Dado $n \in \mathbb{N}$ se considera la matriz $A \in M_{n \times n}$:

$$A = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 & 1 & 2 \\ 1 & 1 & 1 & \dots & 1 & 2 & 2 \\ 1 & 1 & 1 & \dots & 2 & 2 & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 1 & 1 & 2 & \dots & 2 & 2 & 2 \\ 1 & 2 & 2 & \dots & 2 & 2 & 2 \\ 2 & 2 & 2 & \dots & 2 & 2 & 2 \end{pmatrix}$$

- (i) Para $n = 4$ hallar $\det(A)$.
 - (ii) Hallar $\det(A)$ para cualquier n .
-

2.20

Dados $n \in \mathbb{N}$ y $a, b \in \mathbb{R}$ se considera la matriz $A \in M_{n \times n}(\mathbb{R})$:

$$A = \begin{pmatrix} a+b & a & a & \dots & a & a \\ a & a+b & a & \dots & a & a \\ a & a & a+b & \dots & a & a \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a & a & a & \dots & a+b & a \\ a & a & a & \dots & a & a+b \end{pmatrix}$$

- (i) Hallar $\det(A)$ en función de a, b, n .
 - (ii) Hallar $\text{rango}(A)$ en función de a, b, n .
-